

Mathematical Foundations: Complete Reference Index

A Unified Guide to All Theory Documents

Overview

This collection documents the mathematical theories explored across Gaussian Process modelling, Bayesian optimization, multi-output learning, non-stationary processes, numerical linear algebra, dynamical systems, and concurrent systems design. Each document is self-contained but deeply connected to the others. This index describes what each document covers, the key results it contains, and how it relates to the rest of the collection.

Document 1 — Gaussian Process Regression

File: `01_gaussian_process_regression.md`

The foundational document for the entire collection. Introduces the GP as a probability distribution over functions rather than values, defines the mean and kernel functions, and derives the posterior conditioning equations from first principles. Covers the Matérn family in detail — why Matérn 5/2 sits in the sweet spot between excessive smoothness (squared exponential) and excessive roughness (Matérn 1/2) — and explains the role of the kernel as the complete encoding of prior belief about function structure.

Key results include the closed-form GP posterior mean and covariance, the marginal likelihood as a principled objective for hyperparameter optimization, and the log marginal likelihood decomposition into data-fit and complexity-penalty terms that implements an automatic Occam's razor. Closes with a survey of sparse approximations (inducing points, random Fourier features, Kronecker structure) for scaling beyond $O(n^3)$.

Connects to: Document 2 (Cholesky provides the numerical engine for GP inference), Document 3 (acquisition functions consume the GP posterior), Document 4 (multi-output extension), Document 5 (non-stationary extensions), Document 6 (two-timescale hyperparameter optimization).

Document 2 — Cholesky Decomposition and Numerical Linear Algebra

File: 02_cholesky_decomposition.md

The numerical backbone of all GP computations. Derives the Cholesky decomposition $A = LL^T$ from first principles, explains why it exists (positive definiteness), why it is unique (positive diagonal constraint), and how it is computed (column-by-column elimination). Shows how forward and backward substitution turn the decomposition into an efficient and numerically stable linear system solver.

Key results include the log-determinant formula $\log \det(A) = 2 \sum_i \log L_{ii}$ — free once L is available — and the complete GP posterior inference pipeline written entirely in terms of the Cholesky factor. The document also covers the Cholesky parametrization of positive definite matrices ($B = L_B L_B^T$), which is the foundation for learning the coregionalization matrix without constrained optimization, and explains why the jitter term is both practically necessary and statistically principled.

Connects to: Document 1 (provides the solver for GP inference), Document 4 (Cholesky parametrization of B), Document 6 (log-determinant appears in marginal likelihood).

Document 3 — Acquisition Functions

File: 03_acquisition_functions.md

Treats the exploration-exploitation dilemma as the central problem of sequential decision-making, then develops three acquisition functions as progressively more sophisticated solutions. UCB (Upper Confidence Bound) is derived as an optimism-under-uncertainty heuristic with sublinear regret guarantees. Expected Improvement (EI) is derived in full detail from first principles — from the integral definition, through the substitution $z = (y - \mu)/\sigma$, to the closed-form $\text{EI}(x) = (\mu - f^*)\Phi(Z) + \sigma\phi(Z)$ — with geometric interpretations of both the exploitation and exploration terms. Thompson Sampling is presented as the philosophically cleanest approach: draw a random function from the posterior, find its maximum.

The document also covers EI collapse and the sliding window best as the fix, the softmax policy for discrete arm selection, and the theoretical regret bounds that place the three acquisition functions in relation to each other.

Connects to: Document 1 (acquisition functions consume GP posterior), Document 4 (joint

Thompson sampling generalizes to multi-output case).

Document 4 — Multi-Output GP and Coregionalization

File: 04_multi_output_gp_coregionalization.md

Extends the single-output GP to the multi-task setting. Derives the Linear Model of Coregionalization (LMC) from the principle of sharing latent functions across tasks, then specializes to the Intrinsic Coregionalization Model (ICM) where a single shared input kernel is scaled by the coregionalization matrix B . Shows that the resulting joint kernel $K_{\text{joint}} = B \otimes K_{\text{input}}$ has Kronecker structure, and explains the dramatic computational savings this enables: $O(T^3 + n^3)$ instead of $O(T^3 n^3)$.

Derives the joint posterior mean and variance for querying any task given all tasks' observations, which is where the "borrowing strength" property is mathematically expressed. Develops joint Thompson Sampling over the full $(T \cdot G)$ -dimensional posterior — showing why independent samples per task discard the correlation information and why the coherent joint draw preserves it. Covers all three identifiability issues: rotation ambiguity (fixed by positive diagonal), scale ambiguity (fixed by normalization), and ordering ambiguity (absent in the ICM). Closes with coverage-based regularization and the spatio-temporal kernel extension.

Connects to: Documents 1 and 2 (inherits GP inference and Cholesky machinery), Document 3 (joint Thompson sampling extends acquisition functions), Document 5 (temporal component of spatio-temporal kernel), Document 6 (joint hyperparameter optimization including B).

Document 5 — Non-Stationary Gaussian Processes and Spectral Methods

File: 05_non_stationary_gp_spectral.md

Addresses the fundamental limitation of stationary kernels — the assumption that function properties are the same everywhere — and develops the theory for kernels that can represent spatially varying structure. Begins with Bochner's theorem, which establishes the bijection between stationary kernels and spectral densities via the Fourier transform, and derives the spectral densities of the Matérn and squared exponential kernels as concrete examples.

Develops three approaches to non-stationarity: input warping (monotone transformation of

the input space, computationally free but limited); the Gibbs kernel (locally varying length scale, flexible but requires specifying $\ell(x)$); and Deep Kernel Learning (neural network feature map before the kernel, most flexible but most prone to overfitting). Introduces Random Fourier Features as a practical algorithm for large-scale kernel approximation. Closes with locally stationary processes and the time-decay approximation for practical non-stationarity in temporal settings.

Connects to: Document 1 (extends kernel theory), Document 4 (spatio-temporal kernels combine spatial Matérn with temporal periodic kernel), Document 6 (non-stationarity complicates hyperparameter learning).

Document 6 — The Two-Timescale Bayesian Update Problem

File: `06_two_timescale_bayesian_update.md`

Identifies and resolves the asymmetry between function-value updates (fast: one new data point is immediately informative) and hyperparameter updates (slow: global structure requires many observations across the input space). Formulates the hierarchical Bayesian model that underlies the two-timescale approach and explains why fully Bayesian inference over hyperparameters is intractable, motivating the Empirical Bayes approximation (maximize marginal likelihood to get a point estimate $\hat{\theta}$).

Develops the Robbins-Monro framework for stochastic approximation and shows how it governs the convergence of two-timescale algorithms. Derives the logarithmic optimization schedule from information-theoretic considerations about diminishing returns to hyperparameter estimation. Explains the overfitting risk (optimization against the same data used for inference) and the MAP hyperparameter estimation fix. Closes with a concrete implementation architecture separating fast and slow code paths with distinct concurrency responsibilities.

Connects to: Documents 1 and 4 (provides the optimization procedure for all hyperparameters), Document 7 (the two-timescale architecture requires careful concurrent design).

Document 7 — Concurrency Design Patterns

File: `07_concurrency_design_patterns.md`

Treats concurrency not as an implementation detail but as a formal design problem. Introduces the three fundamental hazard classes (races, deadlocks, livelocks), then

develops the lock hierarchy theorem: a system is deadlock-free if and only if the lock acquisition graph is acyclic, and a strict total ordering on locks guarantees acyclicity. Shows how to enforce this at runtime via a hierarchical lock wrapper that converts structural violations into immediate exceptions.

Develops the snapshot pattern (copy state under lock, compute outside lock) from the theoretical principle of immutability as a concurrency primitive. Presents the producer-consumer pattern and message passing as an alternative to shared memory that eliminates races by construction, grounding it in Hoare's CSP formalism. Concludes with concurrent phase annotation as a design tool — making the lock protocol explicit in the code structure rather than leaving it implicit — and a survey of testing strategies for concurrent systems.

Connects to: Document 6 (the two-timescale optimizer requires a producer-consumer architecture with a lock hierarchy).

Document 8 — The Kalman Filter

File: 08_kalman_filter.md

Derives the Kalman filter from first principles as the optimal linear estimator for linear-Gaussian state-space models. Develops the prediction and update equations, explains the Kalman gain as a principled weighting between prior uncertainty and measurement noise, and derives the steady-state behavior. Connects the Kalman filter to Bayesian inference by showing it is exact Bayesian filtering applied to Gaussian distributions.

The document closes with the deep connection between the Kalman filter and the Gaussian Process: stationary GPs can be expressed as state-space models, and GP inference on these models can be performed exactly using the Kalman filter at $O(n)$ cost rather than $O(n^3)$. This is particularly relevant for temporal GPs where observations arrive sequentially. The Matérn 5/2 kernel corresponds to a 3-dimensional state-space model — a concrete link between the kernel theory in Document 1 and the filter theory here.

Connects to: Document 1 (Kalman filter is exact GP inference for temporal models), Document 6 (sequential Bayesian updating relates to the fast timescale of the two-timescale problem).

Document 9 — Chaotic Dynamical Systems and the Logistic Map

File: 09_chaotic_dynamical_systems.md

Develops the mathematics of deterministic chaos through the logistic map $x_{t+1} = rx_t(1 - x_t)$. Traces the bifurcation diagram from stable fixed point through period-doubling cascade to full chaos, introduces Lyapunov exponents as the measure of sensitive dependence on initial conditions, and derives the invariant (arcsine) distribution of the $r = 4$ map. Shows that the map generates exactly $\ln 2$ bits of new information per iteration — the information-theoretic equivalent of a fair coin flip — via the Pesin identity connecting metric entropy to Lyapunov exponents.

Covers the chaotic timing engine as a concrete application: using the logistic map to generate inter-event delays that are statistically unpredictable while requiring only a few arithmetic operations. Derives the mean and variance of the resulting delay distribution analytically from the arcsine measure.

Connects to: Document 3 (timing jitter affects when acquisition function evaluations occur and how the decision loop interacts with the environment), Document 7 (chaotic timing relates to the temporal spacing of events that the concurrent system must handle).

Reading Pathways

Different readers will benefit from different entry sequences through this material.

For readers primarily interested in Bayesian machine learning: Begin with Documents 1 and 2 (GP regression and Cholesky), then 3 (acquisition functions), then 4 (multi-output extension), then 5 (non-stationarity), then 6 (two-timescale learning). Documents 7–9 are supplementary.

For readers primarily interested in concurrent systems: Begin with Document 7 (concurrency patterns), then 6 (two-timescale motivation for the architecture), then skim Documents 1 and 2 for enough GP background to understand what the background optimizer is doing.

For readers interested in the full mathematical picture: Read in document order (1 through 9). Each document builds on the previous ones in a sequence from foundational GP theory through computational methods through system design through supplementary mathematical topics.

Key Unifying Themes

Several mathematical ideas recur across multiple documents and are worth holding in mind as unifying threads.

Positive definiteness appears in Document 1 (kernel matrices must be positive definite for GP inference to be valid), Document 2 (Cholesky decomposition exists iff the matrix is positive definite), Document 4 (coregionalization matrix $\mathbf{B}$ must be positive semidefinite, achieved via Cholesky parametrization), and Document 5 (the Gibbs kernel's prefactor ensures positive definiteness under varying length scales).

The information-computation tradeoff appears in Document 1 (sparse approximations trade accuracy for speed), Document 3 (acquisition functions trade exploration for exploitation), Document 6 (the logarithmic optimization schedule trades adaptation speed for computational cost), and Document 7 (the snapshot pattern trades memory for reduced lock hold time).

Hierarchical structure appears in Document 4 (the LMC hierarchically combines latent functions into task functions), Document 6 (the hierarchical Bayesian model separates function values from hyperparameters), and Document 7 (the lock hierarchy organizes concurrent access into levels with strict ordering).

The Fourier-time duality connects Document 5 (Bochner's theorem: kernels are Fourier transforms of spectral densities) to Document 9 (the logistic map is conjugate to angle-doubling, which is a discrete Fourier doubling map) through the common thread of frequency-domain analysis.